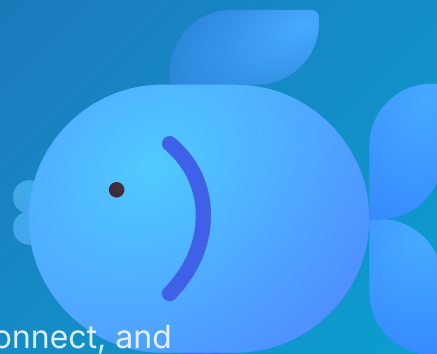


CONFIDENTIAL CLIENT REPORT

InorFresh App Flow Report



A complete explanation of every module, how they connect, and how the entire platform works — written in simple language for easy understanding.

Platform

MERN Stack + Flutter App

Total Modules

6 User Roles

Report Date

June 2026

Prepared By

Development Team

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1. What is InorFresh?

The Big Picture

Simple Explanation

InorFresh is an **online fresh fish & seafood delivery platform** — like Blinkit or Zepto but specialised for fresh fish, seafood, and related products. Customers order from their phone, local sellers pack the order, and a delivery rider delivers it in as little as 10 minutes. The platform also supports B2B bulk fish trade between port sellers and business buyers.



CUSTOMER APP

Shop & Order



SELLER DASHBOARD

**Manage
Stock**



DELIVERY APP

Pick & Drop



ADMIN PANEL

Full Control



2. Who Uses the App?

6 Different User Roles



Customer

Regular shoppers who buy fish and seafood for home or business

- ✓ Browse products by location
- ✓ Add to cart & checkout
- ✓ Track orders live
- ✓ Request returns & refunds
- ✓ Use wallet & coupons



Warehouse / Seller

Local fish shops & suppliers who list products and fulfil orders

- ✓ List & manage products
- ✓ Accept / reject orders
- ✓ Manage inventory & stock
- ✓ View earnings & wallet
- ✓ Handle returns



Delivery Boy

Riders who pick up from sellers and deliver to customers

- ✓ Receive delivery alerts
- ✓ Pick up & deliver orders
- ✓ Verify OTP at delivery
- ✓ Handle return pickups
- ✓ Track earnings



Admin

Platform operators who control everything

- ✓ Approve sellers & products
- ✓ Manage all orders



HORECA / Retailer

Hotels, restaurants, caterers & retail shops buying in bulk

- ✓ Same shopping flow as customer



Port User

Fishing port sellers who list bulk catch for B2B buyers

- ✓ List fish catch / products
- ✓ Post trade requirements

- ✓ Set commissions & fees
- ✓ Send notifications
- ✓ View platform analytics

- ✓ Bulk order capability
- ✓ Separate pricing / accounts
- ✓ Business invoices

- ✓ Negotiate via offers
- ✓ Manage B2B orders



3. Technology Overview

What Powers the App

System Architecture — How All Parts Connect

FRONTEND (WHAT USERS SEE)

 **Flutter Mobile App**

Android & iOS wrapper

 **React Web App**

Customer, Seller,
Delivery, Admin portals

 **Customer Shopping UI**

Product browse, cart,
orders

 **Admin Dashboard**

Full control panel

BACKEND (BRAIN OF THE APP)

 **Node.js + Express**

REST API server

 **Socket.IO**

Real-time updates

 **Firebase Admin**

Push notifications

 **Location APIs**

Geo-based delivery

DATABASE & SERVICES

 **MongoDB Atlas**

All data stored here

 **Razorpay**

Online payments

 **SMS Service**

OTP verification

 **Cloud Storage**

Product images



React + TypeScript

Web UI for all roles



Flutter

Mobile app (Android + iOS)



Node.js + Express

API server & business logic



MongoDB

Database (NoSQL)



Socket.IO



Firebase FCM

Push notifications



Razorpay

Payment gateway



Geo Location

Live order & notification
updates

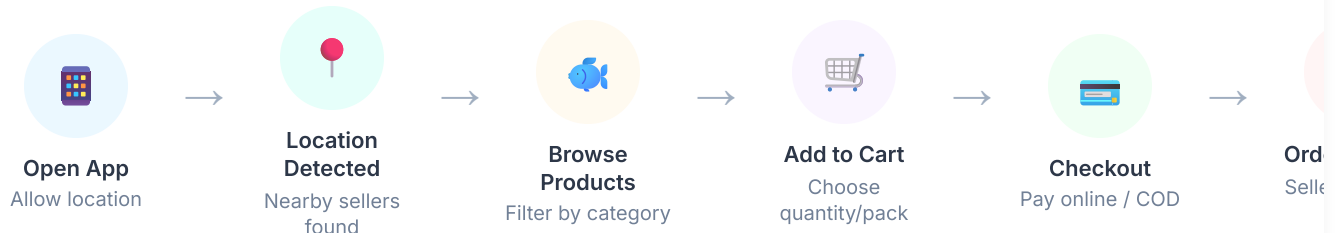
Location-based seller
matching



4. Customer Journey

From App Open to Delivery at Door

Step-by-Step Customer Journey



Registration & Login

1

Enter Phone Number

Customer types their mobile number on the login screen

2

OTP Sent via SMS

A 6-digit OTP is sent immediately to that number

3

Verify OTP

Customer enters OTP — account created or logged in

4

Location Permission

App asks for location to find nearby available sellers



Home Page – Smart Content

- **Location-based products** — only shows items available from sellers near the customer
- **Category browsing** — Fresh Fish, Seafood, Prawns, Cleaned/Cut, etc.
- **Banner promotions** — Admin can push offers, deals, promo strips
- **Best sellers section** — Top selling products highlighted
- **Lowest prices** — Special price section curated by admin
- **Search** — Search any product by name or category
- **Wishlist** — Save favourite products for later



Product Page & Cart

Product Detail

- Product name, weight / pack size
- MRP vs discounted price
- Discount % badge
- Product gallery images
- Description & nutritional info
- Similar products section
- Add to cart / Wishlist button

Checkout

- Select saved address or add new
- Choose payment: Online / COD / Wallet
- Razorpay handles online payments
- Order confirmation with order number
- Push notification sent immediately

Cart

- Add / remove items with + / -
- Apply coupon codes for discount
- See delivery fee + platform fee
- Use wallet balance to pay
- Choose delivery address
- Minimum order quantity enforced

Order Tracking

- Live status updates (Accepted → Packed → On the way → Delivered)
- Delivery boy live tracking on map
- OTP code shown to customer
- Rider verifies OTP at door
- Invoice download after delivery



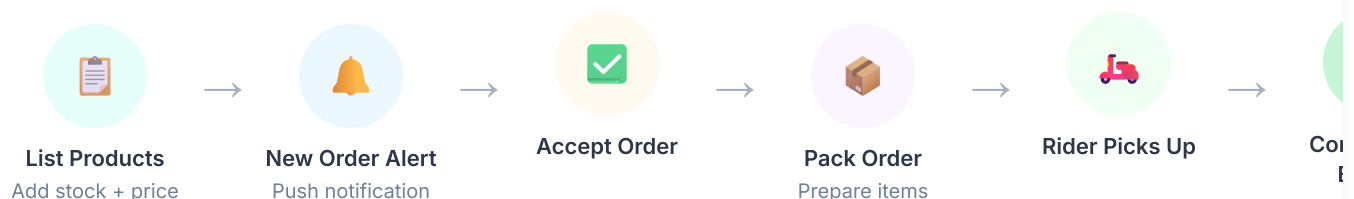
5. Seller / Warehouse Module

How Sellers Manage Their Business

What is a Warehouse/Seller?

A "Warehouse" in InorFresh is a **local fish shop or seafood supplier**. Each seller has a physical location. The platform uses GPS to match a customer's location to nearby sellers — so only relevant products appear to the customer. Sellers set their own product list, prices, and inventory.

Seller Daily Operations Flow



Product Management

- Add new products with photos, price, MRP, and description
- Create product variations (e.g. 500g / 1kg / 2kg packs)
- Set stock quantity — auto goes out-of-stock when 0
- Mark products as active or inactive
- Products need **Admin approval** before going live
- View all products with status (Active / Pending / Rejected)
- Inward stock management — log new stock arrivals

Seller Dashboard & Reports

- Today's orders count & revenue
- Pending / Processing / Delivered order counts
- Monthly revenue graph
- Top selling products
- Wallet balance (money earned)
- Withdrawal request to Admin
- Commission history per order
- Download reports (PDF / CSV)

Location-Based Matching — How It Works

Step 1 – Seller Sets Service Area

Admin assigns each seller a **service radius in KM** (e.g. 5 km). The seller's GPS coordinates are stored.

Step 2 – Customer Opens App

Customer's GPS coordinates are captured. The system finds all sellers within radius.

Step 3 – Products Filtered

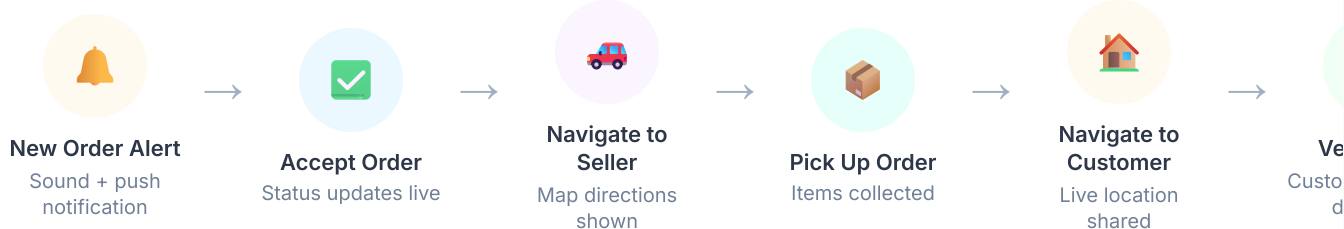
Only products from nearby sellers are shown. Customer never sees out-of-area products.



6. Delivery Boy Module

How Orders Are Picked Up & Delivered

Delivery Boy Order Flow



Key Delivery Features

- **OTP verification** — customer has a permanent 4-digit delivery OTP; rider enters it to confirm delivery
- **Live location tracking** — rider shares GPS; customer sees on map
- **Inspection window** — after delivery, customer has a short time window to inspect and raise return if needed
- **COD collection** — if Cash on Delivery, rider collects cash and logs it
- **Return pickup** — rider picks up returned items from customer and brings back to seller
- **Sellers in range** — rider sees all nearby sellers on map

Delivery Boy Earnings

- Fixed commission per delivery OR percentage-based rate
- Admin sets commission rate for each rider
- Earnings added to wallet after each delivery
- Wallet balance visible in app
- Withdrawal request to Admin
- Cash collected (COD) tracked separately
- Bonus types: Fixed / Salaried / Commission-based



7. Admin Control Panel

Complete Platform Management

What Does the Admin Do?

The Admin is the **super controller of the entire platform**. They approve sellers, approve products, manage all orders, set pricing rules (commission, platform fee, delivery charge), send broadcast notifications, handle disputes, manage wallets, and view analytics for the entire business.

Admin Module Overview

Dashboard

- Total orders today / this month
- Revenue graphs
- Active sellers count
- Active delivery boys
- Platform-level analytics

Product Approval

- Review products submitted by sellers
- Approve or Reject with reason
- Edit product details
- Manage categories & sub-categories
- Set featured / deal-of-day badges

Finance & Payouts

- View platform wallet balance
- Approve withdrawal requests (sellers & riders)
- Set global commission rates
- Platform fee & delivery fee settings
- Tax management (GST config)
- View all payment transactions

User Management

- View all customers
- Add/edit/block sellers
- Add/edit/block delivery boys
- Manage admin team roles & permissions
- View HORECA / Retailer accounts

Order Management

- View all orders across all sellers
- Assign delivery boys to orders
- Override order status if needed
- Handle disputed / escalated orders
- View delivery assignments

App Content

- Home page banners & promo strips
- Best seller cards management
- Lowest prices section
- FAQ & policy pages
- Coupon code creation
- Push notifications to all users



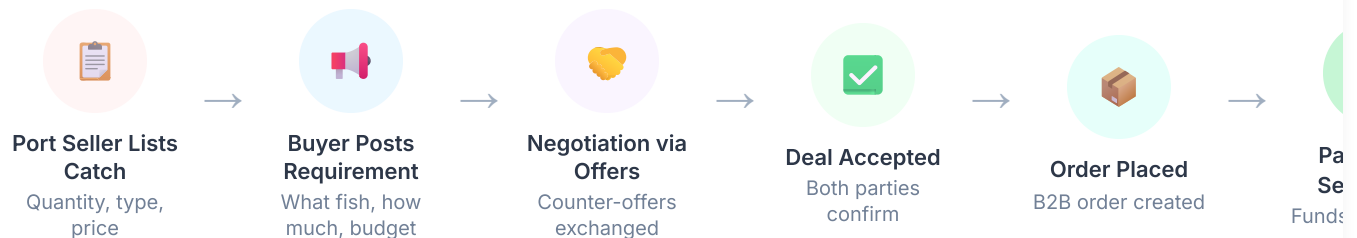
8. Port Module (B2B Trade)

Bulk Fish Trading Between Ports & Businesses

What is the Port Module?

This is a **B2B (business-to-business) trading platform** for fishing ports. Port users are wholesale fish sellers who offload large catches. Business buyers can browse available fish stock, post their requirements, and negotiate prices — similar to a digital fish auction / wholesale market.

Port Trade Flow



9. Complete Order Lifecycle

From "Place Order" to "Delivered" — Every Status Explained

Order Status Timeline

- 1 Received**
Customer places the order. Payment is processed. Seller gets a notification immediately.
- 2 Accepted**
Seller reviews and accepts the order. Customer gets a "Your order is accepted" push notification.
- 3 Processed / Packed**
Seller packs the items and marks order as ready for pickup.
- 4 Picked Up**
Delivery boy picks up from seller. Customer can see rider is on the way.
- 5 On the Way / Out for Delivery**
Rider is heading to the customer. Live map tracking is active.
- 6 Delivered**
Rider enters the customer's OTP at the door. Order marked as delivered. Wallet & commission updated.



Cancelled / Rejected

If seller rejects or customer cancels before pickup, the payment is refunded to customer's wallet.



OTP Delivery System

- Every customer has a permanent 4-digit delivery OTP stored in their account
- Rider sees the OTP on their app at time of delivery
- Customer tells the rider their OTP
- Rider enters it — order marked as Delivered
- This prevents fake delivery claims and protects both customer and rider
- Rider cannot mark delivered without correct OTP



Payment Methods

- **Online Payment** — via Razorpay (UPI, Cards, Net Banking)
- **Cash on Delivery (COD)** — customer pays rider in cash at door
- **Wallet** — use wallet balance (received from refunds or credits)
- **Wallet + Online** — partial payment from wallet, rest online
- Coupon code discount applied at checkout
- Invoice generated after every delivery



10. Return & Refund Flow

What Happens When a Customer Returns an Item

Return Process — Step by Step



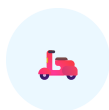
Customer Raises Return

Photo + reason uploaded



Seller Reviews

Approves / Rejects / Escalates



Rider Assigned for Pickup

Return pickup scheduled



Item Collected

OTP verified at pickup



Item Returned to Seller

Warehouse OTP verified



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Return Status Stages

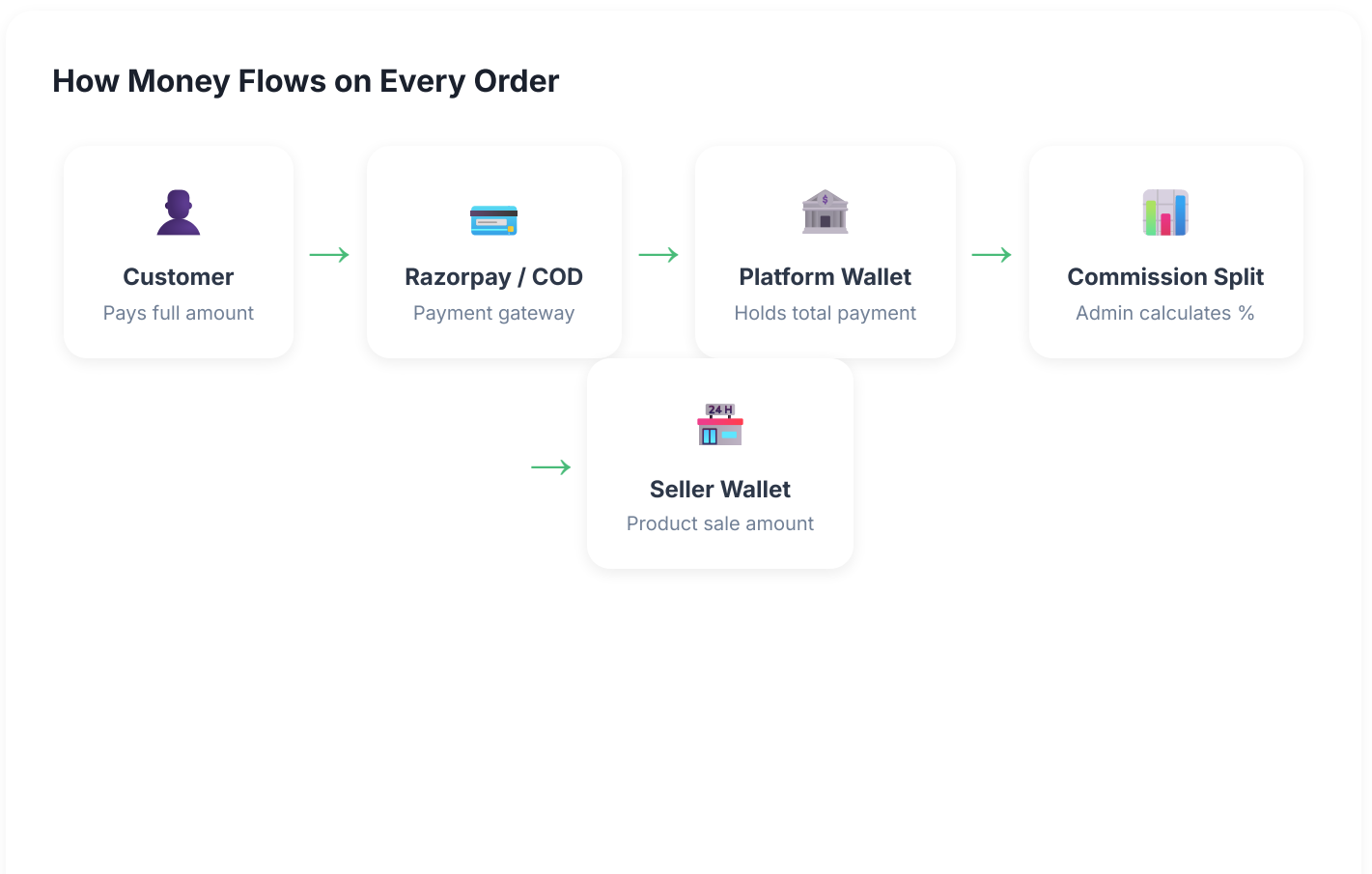
Status	Meaning	Who Acts
--------	---------	----------

REQUESTED	Customer raised a return request with photos	Customer
UNDER REVIEW	Seller is reviewing the request	Seller/Admin
Approved	Return approved, rider to be sent for pickup	Seller
COLLECTED BY RIDER	Delivery boy has picked up the returned item	Rider
IN TRANSIT TO WAREHOUSE	Item being brought back to seller	Rider
RECEIVED AT WAREHOUSE	Seller confirmed receipt with OTP	Seller
REFUND APPROVED	Refund amount confirmed	Admin/Seller
REFUNDED	Money credited to customer wallet	System
Rejected	Return rejected (not eligible)	Seller/Admin



11. Money Flow

How Money Moves Between Platform, Sellers, Riders & Customers





Platform Fee

Stays with InorFresh



Delivery Fee

Paid to delivery boy



Rider Wallet

Per-delivery earnings



Wallet System

- **Customer wallet** — Refunds go here; can be used to pay for orders
- **Seller wallet** — Commission from each delivered order credited here; withdraw to bank via admin
- **Delivery boy wallet** — Per-delivery earnings; withdraw to bank via admin
- **Platform wallet** — Platform fee and delivery charge held here; admin controls payouts
- Every wallet transaction (credit/debit) is logged with opening and closing balance



Commission Structure

- Admin sets a **global commission rate (%)** for all sellers
- Individual sellers can have a different rate (override)
- Delivery boys have their own commission rate per delivery
- Commission is calculated automatically when order is Delivered
- Admin can view all commissions earned per order
- Withdrawal requests from sellers/riders require Admin approval



12. Notifications System

How Everyone Stays Updated in Real-Time

Three Types of Notification Delivery



Push Notification (FCM)

Sent to mobile/browser even when the app is closed. Uses Firebase Cloud Messaging. Appears as a system notification on the phone.



Real-Time (Socket.IO)

Instant in-app alerts while the app is open. When a new order arrives, the seller hears a sound and sees it immediately without refreshing.

In-App Notification Inbox

All notifications are stored and can be read from the Notifications page. Shows read/unread count with a badge.

What Triggers a Notification?

Event	Who Gets Notified	Message
New order placed	Seller + Admin	"New order received! #ORD-1234"
Order accepted by seller	Customer	"Your order has been accepted"
Order packed / ready	Delivery Boy	"New order ready for pickup"
Order picked up	Customer	"Your order is on the way!"
Order delivered	Customer + Seller	"Order delivered successfully"
Order cancelled	Customer	"Your order was cancelled. Refund initiated."
Return request raised	Seller + Admin	"New return request for order #ORD-1234"
Refund processed	Customer	"Refund of ₹X credited to your wallet"
Product approved	Seller	"Your product has been approved"
Admin broadcast	All users	Custom message from Admin



13. Flutter Mobile App

How the Mobile App Works

What is the Flutter App?

The mobile app is built with **Flutter** — Google's framework for building Android and iOS apps. It wraps the InorFresh React web app inside a WebView (like a powerful in-app browser), so

Why This Approach?

Building one web app + wrapping it in Flutter means the team only maintains **one codebase** for web + Android + iOS. Any updates go live on

both the website and app share the same codebase and features. The Flutter layer adds native features like system push notifications, camera access, and app store distribution.

all platforms simultaneously. No separate mobile app code to maintain.

Flutter App Features (Native Layer)

Native Push Notifications

- Uses native Firebase SDK
- Notifications appear in Android/iOS system tray
- Work even when app is fully closed
- Tapping opens the relevant screen

Location Access

- Requests device GPS permission
- Passes location to web app
- Used for seller matching

App Store Distribution

- Published on Google Play Store
- Can be published on Apple App Store
- Users install like any normal app
- Auto-update support

Performance

- App-shell caching (loads instantly)
- Offline fallback page
- Images cached locally
- Smooth transitions



14. Security & Authentication

How the App Keeps Data Safe

Security Measures

- **JWT Tokens** — All API calls require a valid login token that expires and is refreshed
- **OTP Verification** — Phone number verified via SMS OTP at registration and key actions
- **Role-Based Access** — Each user type (Customer, Seller, Rider, Admin) can only access their own data and actions
- **Delivery OTP** — Prevents fake delivery claims — only customer can confirm delivery
- **Password Hashing** — All passwords stored encrypted (bcrypt) — never stored plain text
- **HTTPS only** — All communication between app and server is encrypted
- **Razorpay** — All payments processed by Razorpay (RBI-approved) — InorFresh never stores card details
- **FCM Token dedup** — Push notification tokens are cleaned per device when users switch

→ **Return OTP** — Warehouse verifies receipt of returned items via OTP

accounts

→ **Input Validation** — All inputs validated on backend to prevent injection attacks



Summary — How It All Works Together

The Full Platform in One Diagram



Key Numbers at a Glance



USER ROLES



ORDER STATUSES



RETURN STATUSES



NOTIFICATION
TYPES

6 Types

12 Stages

10 Stages

3 Channels



PAYMENT
METHODS

4 Options



WALLET TYPES

4 Wallets



DATABASE
MODELS

50+ Tables



API ENDPOINTS

100+ APIs

 **InorFresh Platform**

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